Ahmed Aziz Ben Haj Hmida

Machine Learning Engineer, graduate of the École Polytechnique Fédérale de Lausanne (EPFL), with a focus on Computer Vision and Large Language Models in industrial or product-focused environments.

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Education

M.Sc. in Data Science 09/2023 - 03/2026
EPFL, Lausanne, Switzerland
Courses: Machine Learning, Natural Language Processing, Data Analysis, Image Analysis and Pattern Recognition, Reinforcement Learning, Database Systems, Data Science for Business
GPA: 5.3/6

3rd Year of Bachelor - Exchange 08/2022 - 07/2023
Technical University of Denmark, Lyngby, Denmark

B.Sc. in Communication Systems 09/2020 – 07/2023
EPFL, Lausanne, Switzerland
Courses: Signal Processing, Principles of Digital Communication
GPA: 5.1/6

Baccalaureate in Mathematics 09/2018 - 06/2019
Pioneer School of Nabeul, Tunisia

Professional Experience

Master's Thesis: Curve-Based Similarity Detection & Parameter Recalibration for CNC Laser Tools (Grade 6/6) 08/2025 - 03/2026
Company: Rollomatic SA, Le Landeron, Switzerland
Design and deploy a curve-based similarity search system for CNC laser machining tools, combining geometric curve representations and image-based embeddings. Improve tool retrieval accuracy by **+11%** in industrial design workflows. Reduce tool search time by **95–99%, from minutes to seconds**.
Supervisor: Prof. Jean-Philippe Thiran

Internship: Hybrid Similarity Search System for Laser Machining Tools 02/2025 - 08/2025
Company: Rollomatic SA, Le Landeron, Switzerland
Preprocess tool data and develop a prototype retrieval pipeline for tool similarity leveraging parameters and deep visual embeddings.
Supervisor: Dr. Jonathan Rochat

Research Projects

Unsupervised reverse knowledge distillation and model merging 08/2024 - 02/2025
EPFL LTS4 Lab
Transfer knowledge from a small, simple model to a large, more complex one then merge the transferred specialised model into the generalist base model.
Skills: Text/Image data processing, Knowledge Distillation, LLM, Transformers, Model Merging.
Supervisor: Prof. Pascal Frossard

Discovering promising new startups through evaluating teams' potentials 09/2023 - 12/2023
EPFL ENTC Lab
Harness Machine Learning methods to derive speakers' MBTI personalities and emotions during entrepreneurial discussions to identify the most promising teams for creating successful startups.
Skills: Text/Audio data processing, LSTM networks.
Supervisor: Prof. Davide Bavato

Academic Projects

Coin Detector and Counter 02/2024 - 06/2024
Use segmentation techniques and deep learning models (CNN, Vision Transformer, ResNet50) to detect and identify coins accurately across diverse backgrounds.

LLM as a Teaching Assistant: Leveraging LLM for MCQ Answering 02/2024 - 06/2024
Finetune LLMs (Qwen1.5) as teaching assistants for MCQs with Direct Preference Optimization, supervised learning and quantization.

Exploring the transition from novice to expert in RL policies for motor skill 02/2024 - 06/2024
Lab: EPFL Alexander Mathis Lab
Train an expert model on the Myosuite baoding balls task, distill its motor strategies to novice agents to improve learning efficiency and performance of novice agents in complex motor control tasks.

Machine Learning Operations 01/2023 - 03/2023
Deploy ML models using PyTorch, Docker and Cloud services.

Agile Object-Oriented Programming 09/2020 - 05/2023
Create Java applications (Pacman, Ticket to Ride, Battleship), focusing on the agile methodology.

Technical Skills

Programming Languages: Python, Java, SQL, Matlab, Scala

Machine Learning: NumPy, Pandas, scikit-learn, PyTorch, XGBoost, Matplotlib, Seaborn

Computer Vision: OpenCV, torchvision, Pillow, scikit-image

Natural Language Processing: spaCy, NLTK, Hugging Face Transformers, SentenceTransformers, Gensim

Tools & MLOps: Git, Docker, FastAPI, MLflow, W&B, CI/CD (GitHub Actions), Cloud (GCP), Streamlit

Associations

EPFL Coaching - Coach of CMS section 09/2023 - 06/2024

Rotary International - Member 07/2018 - 04/2019

Tunisian Red Crescent - Volunteer 05/2018 - 04/2019

Certificates

Structuring ML Projects 09/2023 - 10/2023
DeepLearning.AI, Coursera

Certificate in Artificial Intelligence 06/2023 - 09/2023
GOMYCODE

Languages

Arabic: Native

French: Native

English: Professional Proficiency (C1)

German: Limited Proficiency (A2)